

SUPPORT VECTOR MACHINE (SVM)

Support Vector Machine (SVM) is a **Supervised Machine Learning Algorithm** mainly used for:

-  Classification
-  Regression (called SVR – Support Vector Regression)

It is one of the most powerful algorithms for **classification problems**.

WHAT DOES SVM DO?

SVM tries to find the **best possible boundary (called a hyperplane)** that separates different classes.

Imagine you have two groups of objects:

 Apples

 Oranges

SVM draws the **best line** between them so that future fruits can be classified correctly.

Example:

Apple Apple

| Best Boundary |

Orange Orange

The vertical line is called the **Decision Boundary** or **Hyperplane**.

REAL-LIFE EXAMPLE

Suppose a bank wants to decide whether to approve a loan.

Data:

Salary Loan Status

₹20,000 Rejected

₹25,000 Rejected

Salary Loan Status

₹70,000 Approved

₹80,000 Approved

Now a customer with ₹60,000 salary applies.

SVM finds the best boundary and predicts whether the loan should be **Approved** or **Rejected**.

EXAMPLE 1: STUDENT PASS/FAIL

Training Data

Study Hours Result

2 Fail

3 Fail

4 Fail

6 Pass

7 Pass

8 Pass

Graphically

Fail Fail Fail Pass Pass Pass

2 3 4 | 6 7 8

↑

Best Boundary

If a student studies **5 hours**, SVM predicts based on which side of the boundary the point falls.

WHAT IS A HYPERPLANE?

A **hyperplane** is the boundary that separates different classes.

- In **2D**, it is a **line**.
- In **3D**, it is a **plane**.
- In higher dimensions, it is called a **hyperplane**.

Example:

o o o o

x x x x

The horizontal line is the hyperplane.

WHAT ARE SUPPORT VECTORS?

Support vectors are the **data points closest to the boundary**.

Example:

o o o o

o

x

x x x x

The nearest o and x are the **support vectors**.

These points determine where the boundary should be placed.

WHAT IS MARGIN?

The **margin** is the distance between the hyperplane and the nearest support vectors.

o o

===== Margin =====

----- Hyperplane

===== Margin =====

x x

SVM tries to **maximize the margin**.

A larger margin usually leads to better generalization.

HOW SVM WORKS

Step 1

Receive training data.

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Step 2

Find all possible separating lines.

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Step 3

Calculate the margin for each line.

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Step 4

Choose the line with the **largest margin**.

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Step 5

Use that boundary to classify new data.

TYPES OF SVM

1. Linear SVM

Used when the data can be separated by a straight line.

Example:

Apple Apple

Orange Orange

2. NON-LINEAR SVM

Used when a straight line cannot separate the data.

Example:

o o o

o o

x

o o

o o o

A curved boundary is needed.

KERNEL TRICK

The **Kernel Trick** allows SVM to handle non-linear data by transforming it into a higher-dimensional space where it becomes easier to separate.

Common kernels:

Kernel	Use
Linear	Linear data
Polynomial	Curved boundaries
RBF (Radial Basis Function)	Most common for complex data
Sigmoid	Similar to neural networks

ADVANTAGES OF SVM

- Works well on small and medium-sized datasets.
 - Effective when there are many features (high-dimensional data).
 - Finds a robust decision boundary by maximizing the margin.
 - Can model complex patterns using kernels.
 - Often performs well on text classification problems.
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DISADVANTAGES OF SVM

- Training can be slow on very large datasets.
 - Choosing the right kernel and parameters can be difficult.
 - Sensitive to noisy or overlapping data.
 - Does not naturally provide probability estimates unless configured to do so.
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WHEN SHOULD WE USE SVM?

- ✓ Email spam detection
- ✓ Face recognition
- ✓ Image classification
- ✓ Text classification
- ✓ Handwriting recognition
- ✓ Medical diagnosis

- ✔ Gene or DNA classification
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WHEN SHOULD WE NOT USE SVM?

- ✗ Very large datasets (hundreds of thousands or millions of records)
 - ✗ Data with many missing values (clean the data first)
 - ✗ When fast training is more important than accuracy
 - ✗ When the classes overlap heavily and are difficult to separate
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SVM vs KNN

Feature	SVM	KNN
Learning Type	Supervised	Supervised
Training	Builds a decision boundary	Stores the training data
Prediction Speed	Fast after training	Slower on large datasets
Large Datasets	Better than KNN for prediction speed, but training can be slow	Prediction becomes slow
Feature Scaling	Recommended	Essential
Best For	Complex classification	Small datasets with similarity-based decisions

INTERVIEW QUESTIONS

1. What is SVM?

A supervised machine learning algorithm that finds the **best separating boundary (hyperplane)** between classes.

2. Is SVM used for classification or regression?

Both. It is mainly used for **classification**, but it also supports **regression** through **Support Vector Regression (SVR)**.

3. What is a hyperplane?

A boundary that separates different classes.

4. What are support vectors?

The training data points closest to the decision boundary. They determine the position of the hyperplane.

5. What is the kernel trick?

A technique that enables SVM to separate non-linear data by mapping it into a higher-dimensional space.

6. Why is feature scaling important in SVM?

SVM relies on distances and optimization. Features with much larger numeric ranges can dominate the model, so scaling (using methods such as StandardScaler) is generally recommended.

SUMMARY

Support Vector Machine (SVM) is a supervised learning algorithm that classifies data by finding the optimal decision boundary (hyperplane) with the maximum margin between different classes, leading to strong generalization on unseen data.